

DATA471 Assignment 4

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Exercise 1

Regarding the computation of order statistics

Hyndman and Fan (1998) proposed 9 ways to compute the quantile over a set of sample data.

Generally speaking, to obtain the quantile p for ordered data x of size N , the function will look at $x[\text{floor}(p*N)]$ and $x[\text{floor}(p*N) + 1]$, and use one of two general methods to compute the quantile:

1. a discrete method that picks a value between these two ordered values in x set by another constant
2. a continuous interpolation method that “smooths” the quantile values when $x[\text{floor}(p*N)] <> x[\text{floor}(p*N) + 1]$

Extreme values of p (very close to 0 or 1) are handled explicitly to be $x[1]$ and $x[N]$ respectively.

Wicklin shows that for $n \geq 100$, the difference between the methods are minimal. Differences are noticeable for smaller datasets however, and so in these scenarios it’s important to report which one you used.

My version of R uses `type=7`, which is one of the continuous interpolation methods.

I would use the default unless there was strong reasoning for otherwise (e.g. small dataset where a discrete method is more appropriate given the context). Using the default is common practice and therefore makes it quicker simpler for other statisticians to understand outputs.

First-order bootstrap bias corrected estimator

First, we create the function that allows us to compute our original estimator.

```
lambda_iqr <- function(data){  
  return (log(3) / IQR(data))  
}  
  
# an example:  
x <- rexp(10, rate=1)  
lambda_iqr(x)
```

```
## [1] 1.378568
```

We then create a function that uses bootstrapping to estimated the expected value.

```
bootstrap_expected_value <- function(data){
  n <- 200
  boot_output <- rep(0, n)

  for(l in 1:n) {
    boot_output[l] <- lambda_iqr(sample(data, size=length(data), replace=TRUE))
  }
  # print(bootstrapped_lambda)
  return(mean(boot_output))
}
# an example:
bootstrap_expected_value(x)
```

```
## [1] 1.71672
```

Lastly, create the function that returns the bootstrap bias corrected estimator for λ_{IQR} :

```
lambda_tilda <- function(data){
  (2*lambda_iqr(data)) - bootstrap_expected_value(data)
}
```

Checking it works:

```
x <- rexp(10, rate=1)
paste0("Lambda IQR: ", lambda_iqr(x))
```

```
## [1] "Lambda IQR: 0.907758861679703"
```

```
paste0("Lambda Tilda: ", lambda_tilda(x))
```

```
## [1] "Lambda Tilda: 0.382531915571352"
```